

**MICROCONTROLLER SYSTEM
AND DESIGN
FINAL PROJECT REPORT**

**FINGERPRINT
SECURITY
SYSTEM**

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PROJECT FINAL REPORT

- **ABSTRACT:**

Human detection and recognition are very important and has developed quickly over time. Fingerprint identification is a crucial and highly trustworthy human identifying technique. Fingerprint of every individual is special. As a result, this aids in system security improvement or individual identification.

By preventing unwanted access and keeping track of who passes through, this security system will offer sufficient security. Sometimes unauthorized people or burglars try to break the lock for evil intentions at a time when no one is available at a targeted place. Password-based security systems and RFID-based security systems have both been seen. The primary quality or specialty of a fingerprint is its singularity. In comparison to other security systems, it gives this project a high level of security.

- **OBJECTIVE**

Establishing a system aimed at providing protection, safety and reduce exposure to common hazards and alert as soon as they occur through fingerprint identification of the person.

- **INTRODUCTION**

The main aim of this project is to develop a security lock system based on fingerprint scanning. Fingerprint sensor is the main part of this system. It is also called as Biometric sensor. User has to place his/her finger on the optical sensor part of fingerprint module.

The concept for this project is that first we have to operate this project in “**Privilege mode or Admin mode**”. In this mode, we have to enter data into the database of fingerprint sensors, and for this, we have to take impressions of the fingerprints of the person to whom we want to give access to our security system. This can be done once or whenever a new entry has to be added in the system.

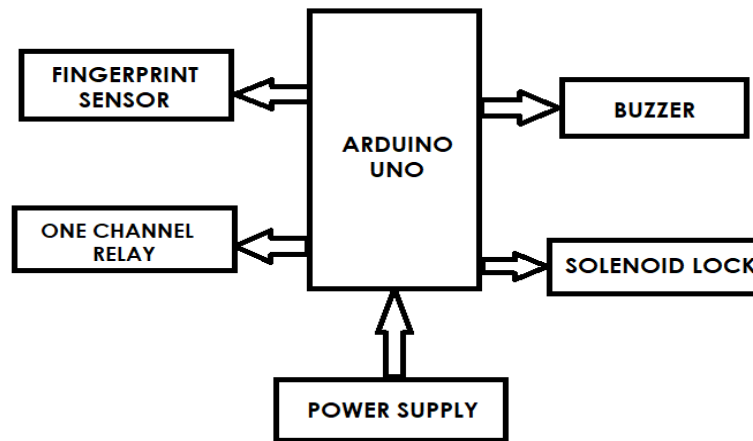
Then this project has to be used in “**Normal mode or user mode**”. In this mode the system compares the fingerprint input received at its optical plate with the previously stored fingerprint from its flash memory. If the entry matches with the memory, then it gives out ok signal. But if the entry does not match with the memory then it gives out error signal. Thus we ensure a secure fingerprint authorized security system.

- **LIST OF COMPONENTS**

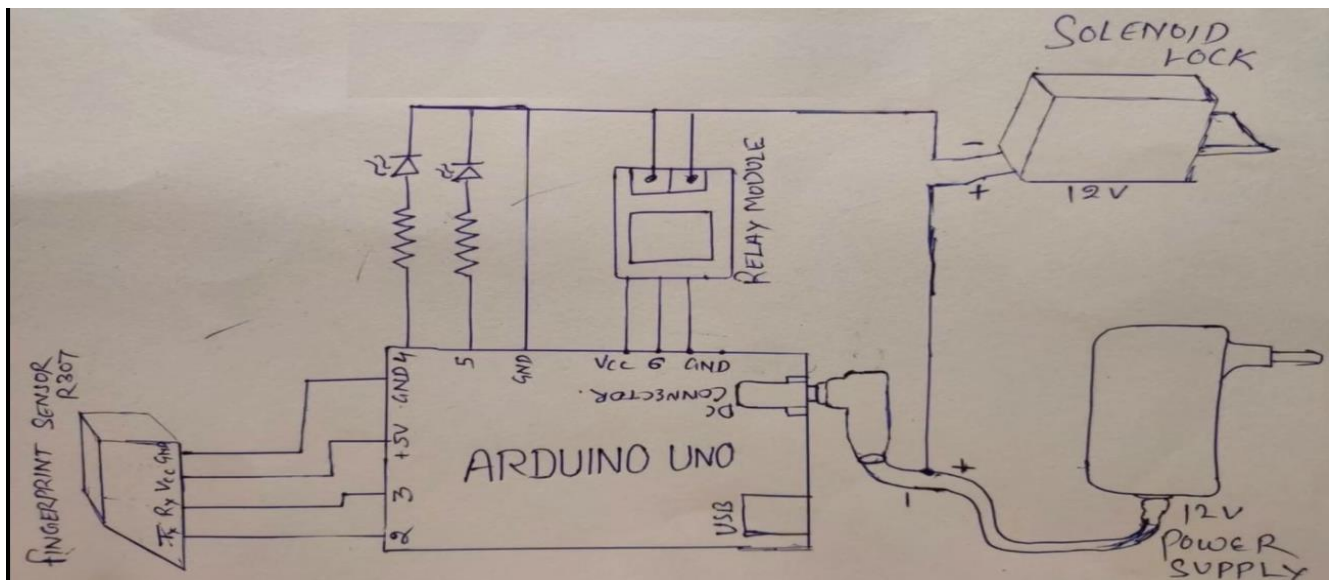
1. Adaptor 12V
2. Arduino UNO
3. Fingerprint Sensor
4. Buzzer
5. Jumper Wires
6. One channel relay
7. Solenoid Lock

- REPRESENTATION

- BLOCK DIAGRAM



- CIRCUIT DIAGRAM



- CONNECTIONS:

The fingerprint Module is directly connected to pin 2 and 3 of Arduino UNO with respect to ground, buzzer is connected at the Digital pin 8 of Arduino UNO and with respect to the ground. Whereas, the relay is connected with the pin 9 with solenoid lock.

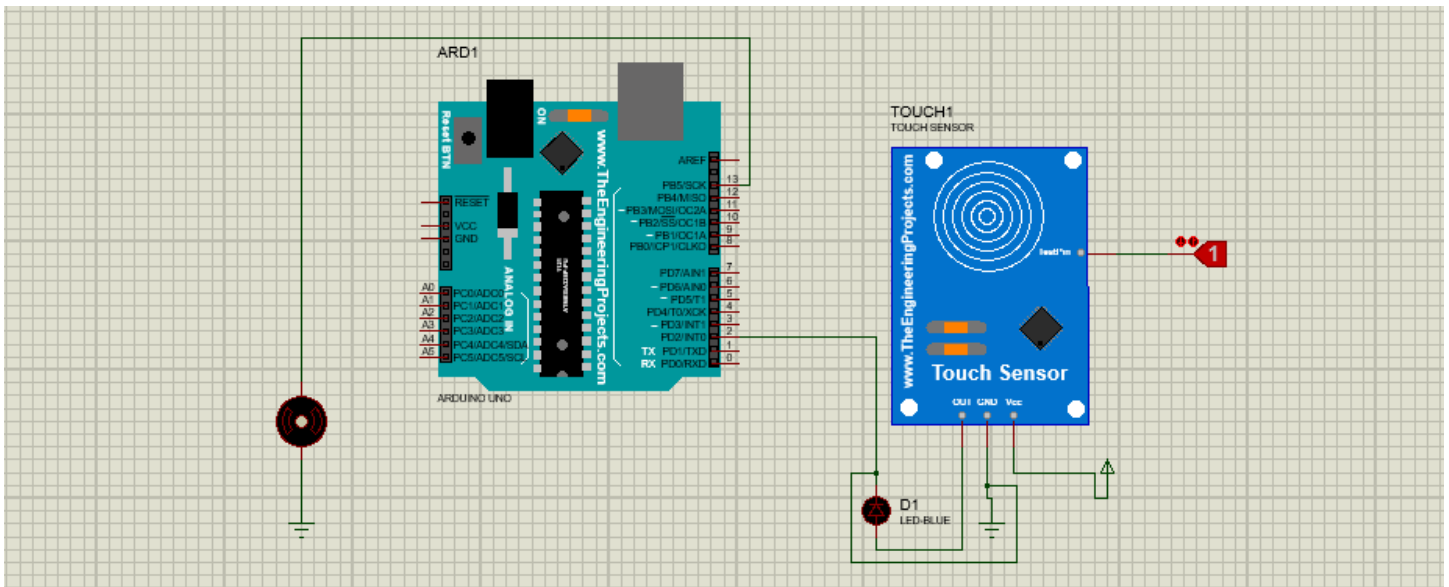
- WORKING

1. The circuit of this fingerprint-based door lock system contains Arduino UNO which controls whole the process of the project buzzer, and solenoid lock.
2. We have used different Arduino codes for performing certain steps regarding the security measures. For example firstly, "**ENROLL**" code is used to enroll the fingerprint of the authenticate user or we can say the admin and that data is stored into the database of

Arduino. For the display purpose, we have used the serial monitor instead of the LCD because for that we have to use resistors that will overall affect the contrast of LCD and will also make the circuit more complex.

3. Then in the next step, **FINGERPRINT MATCH** code is used to match the data stored in the database with the fingerprint data of the user, whether it is the same as that of the admin or not. When any user tries to open the gate, module checks the authenticity of the user by comparing his/her fingerprint with the database, if the users fingerprint matches with the one that is stored in the database, then Arduino sends the signal to open the door lock.
4. On the other hand, if the fingerprint of the user doesn't matches with the one of admin, it means there is any intruder trying to enter, and for this the buzzer at the output port will start ringing, and the door will not open.
5. The solenoid lock used in our project is of 12V, so in order to operate it we have used a 12V adaptor, otherwise to operate the Arduino and door lock with the same voltage, we can use a 9V battery connected with capacitors to distribute the flow of current.

- **PROTEUS SIMULATION**



- **SOURCE CODE**

ENROLL CODE:

```
void loop()
{
  Serial.println("Ready to enroll a fingerprint!");
  Serial.println("Please type in the ID # (from 1 to 127) you want to save this finger as...");
  id = readnumber();
  if (id == 0) { // ID #0 not allowed, try again!
```

```

return;
}
Serial.print("Enrolling ID #");
Serial.println(id);

while (! getFingerprintEnroll() );
}

uint8_t getFingerprintEnroll() {

    int p = -1;
    Serial.print("Waiting for valid finger to enroll as #"); Serial.println(id);
    while (p != FINGERPRINT_OK) {
        p = finger.getImage();
        switch (p) {
            case FINGERPRINT_OK:
                Serial.println("Image taken");
                break;
            case FINGERPRINT_NOFINGER:
                Serial.println(".");
                break;
            case FINGERPRINT_PACKETRECEIVEERR:
                Serial.println("Communication error");
                break;
            case FINGERPRINT_IMAGEFAIL:
                Serial.println("Imaging error");
                break;
            default:
                Serial.println("Unknown error");
                break;
        }
    }

    // OK success!
    p = finger.image2Tz(1);
    switch (p) {
        case FINGERPRINT_OK:
            Serial.println("Image converted");
            break;
        case FINGERPRINT_IMAGEMESS:
            Serial.println("Image too messy");
            return p;
        case FINGERPRINT_PACKETRECEIVEERR:
            Serial.println("Communication error");
            return p;
        case FINGERPRINT_FEATUREFAIL:
            Serial.println("Could not find fingerprint features");
            return p;
        case FINGERPRINT_INVALIDIMAGE:
            Serial.println("Could not find fingerprint features");

```

```

        return p;
    default:
        Serial.println("Unknown error");
        return p;
    }

    Serial.println("Remove finger");
    delay(2000);
    p = 0;
    while (p != FINGERPRINT_NOFINGER) {
        p = finger.getImage();
    }
    Serial.print("ID "); Serial.println(id);
    p = -1;
    Serial.println("Place same finger again");
    while (p != FINGERPRINT_OK) {
        p = finger.getImage();
        switch (p) {
            case FINGERPRINT_OK:
                Serial.println("Image taken");
                break;
            case FINGERPRINT_NOFINGER:
                Serial.print(".");
                break;
            case FINGERPRINT_PACKETRECEIVEERR:
                Serial.println("Communication error");
                break;
            case FINGERPRINT_IMAGEFAIL:
                Serial.println("Imaging error");
                break;
            default:
                Serial.println("Unknown error");
                break;
        }
    }

    // OK converted!
    Serial.print("Creating model for #"); Serial.println(id);
    p = finger.createModel();
    if (p == FINGERPRINT_OK) {
        Serial.println("Prints matched!");
    } else if (p == FINGERPRINT_PACKETRECEIVEERR) {
        Serial.println("Communication error");
        return p;
    } else if (p == FINGERPRINT_ENROLLMISMATCH) {
        Serial.println("Fingerprints did not match");
        return p;
    } else {
        Serial.println("Unknown error");
        return p;
    }
}

```

```

Serial.print("ID "); Serial.println(id);
p = finger.storeModel(id);
if (p == FINGERPRINT_OK) {
    Serial.println("Stored!");
} else if (p == FINGERPRINT_PACKETRECEIVEERR) {
    Serial.println("Communication error");
    return p;
} else if (p == FINGERPRINT_BADLOCATION) {
    Serial.println("Could not store in that location");
    return p;
} else if (p == FINGERPRINT_FLASHERR) {
    Serial.println("Error writing to flash");
    return p;
} else {
    Serial.println("Unknown error");
    return p;
}
return true;
}

```

This is enroll code here we stored the finger which we want to give the access to our Home. We have stored this finger with the ID# 26

```

.
Image taken
Image converted
Remove finger
ID 26
Place same finger again
.....Image taken
Image converted
Creating model for #26
Prints matched!
ID 26
Stored!
Ready to enroll a fingerprint!
Please type in the ID # (from 1 to 127) you want to save this finger as...

```

FINGERPRINT and LOCK CODE:

```

#include <Adafruit_Fingerprint.h>
#if (defined(__AVR__) || defined(ESP8266)) && !defined(__AVR_ATmega2560__)
SoftwareSerial mySerial(2, 3);
#else
#define mySerial Serial1
#endif
Adafruit_Fingerprint finger = Adafruit_Fingerprint(&mySerial);
void setup()
{
    pinMode(9,OUTPUT);//FOR DOOR_LOCK
    digitalWrite(9,HIGH);
    pinMode(8,OUTPUT);//FOR BUZZER
    Serial.begin(9600);
}

```

```

while (!Serial);
delay(100);
Serial.println("\n\nAdafruit finger detect test");
// set the data rate for the sensor serial port
finger.begin(57600);
delay(5);
//SENSING THE FINGERPRINT MODULE
if (finger.verifyPassword()) {
    Serial.println("Found fingerprint sensor!");
} else {
    Serial.println("Did not find fingerprint sensor :(");
    while (1) { delay(1); }
}

Serial.println(F("Reading sensor parameters"));
finger.getParameters();
Serial.print(F("Status: 0x")); Serial.println(finger.status_reg, HEX);
Serial.print(F("Sys ID: 0x")); Serial.println(finger.system_id, HEX);
Serial.print(F("Capacity: ")); Serial.println(finger.capacity);
Serial.print(F("Security level: ")); Serial.println(finger.security_level);
Serial.print(F("Device address: ")); Serial.println(finger.device_addr, HEX);
Serial.print(F("Packet len: ")); Serial.println(finger.packet_len);
Serial.print(F("Baud rate: ")); Serial.println(finger.baud_rate);
finger.getTemplateCount();

if (finger.templateCount == 0) {
    Serial.print("Sensor doesn't contain any fingerprint data. Please run the 'enroll'
example.");
}
else {
    Serial.println("Waiting for valid finger...");
    Serial.print("Sensor contains "); Serial.print(finger.templateCount); Serial.println("
templates");
}
}

void loop() // run over and over again
{
    getFingerprintID();
    delay(50);
}

/**THIS IS THE FINGERPRINT TAKING AND ITS DETECTION PART***/
uint8_t getFingerprintID() {
    uint8_t p = finger.getImage();
    switch (p) {
        case FINGERPRINT_OK:
            Serial.println("Image taken");
            break;
        case FINGERPRINT_NOFINGER:
            Serial.println("No finger detected");
            return p;
    }
}

```



```

case FINGERPRINT_PACKETRECEIVEERR:
    Serial.println("Communication error");
    return p;
case FINGERPRINT_IMAGEFAIL:
    Serial.println("Imaging error");
    return p;
default:
    Serial.println("Unknown error");
    return p;
}
// ***OK finger is Detected Successfully!***
//***This part is for the conversion of taken image of finger***
p = finger.image2Tz();
switch (p) {
case FINGERPRINT_OK:
    Serial.println("Image converted");
    break;
case FINGERPRINT_IMAGEMESS:
    Serial.println("Image too messy");
    return p;
case FINGERPRINT_PACKETRECEIVEERR:
    Serial.println("Communication error");
    return p;
case FINGERPRINT_FEATUREFAIL:
    Serial.println("Could not find fingerprint features");
    return p;
case FINGERPRINT_INVALIDIMAGE:
    Serial.println("Could not find fingerprint features");
    return p;
default:
    Serial.println("Unknown error");
    return p;
}
// ***Now Successfully converted!***
//***Now finally we get our result either match is found or not***
p = finger.fingerSearch();
if (p == FINGERPRINT_OK) {
    Serial.println("Found a print match!");
    digitalWrite(9,LOW);//LOCK OPEN
    delay(2000);
    digitalWrite(9,HIGH);//LOCK CLOSE AFTER SOME DELAY
}
else if (p == FINGERPRINT_PACKETRECEIVEERR) {
    Serial.println("Communication error");
    return p;
} else if (p == FINGERPRINT_NOTFOUND) {
    Serial.println("Did not find a match");
    for (int b = 0; b <= 1500; b = b+500 ){
        digitalWrite(8,HIGH);//BUZZER WILL BEEP
        delay(250);
    }
}

```

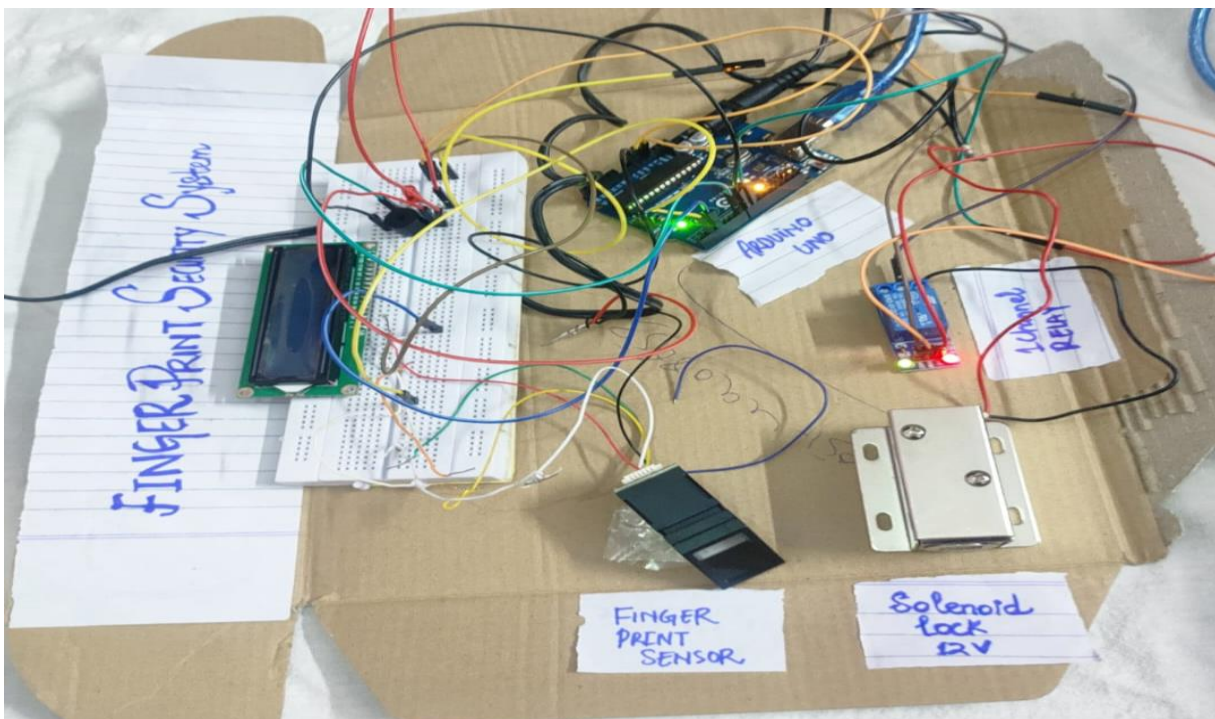
```

    digitalWrite(8,LOW);//DOOR LOCK WILL NOT OPEN
    delay(250);
}
    return p;
} else {
    Serial.println("Unknown error");
    return p;
}
// found a match!
Serial.print("Found ID #"); Serial.print(finger.fingerID);
Serial.print(" with confidence of "); Serial.println(finger.confidence);
return finger.fingerID;
}
// returns -1 if failed, otherwise returns ID #
int getFingerprintIDez() {
    uint8_t p = finger.getImage();
    if (p != FINGERPRINT_OK) return -1;
    p = finger.image2Tz();
    if (p != FINGERPRINT_OK) return -1;
    p = finger.fingerFastSearch();
    if (p != FINGERPRINT_OK) return -1;
    // found a match!
    Serial.print("Found ID #"); Serial.print(finger.fingerID);
    Serial.print(" with confidence of "); Serial.println(finger.confidence);
    return finger.fingerID;
}

```

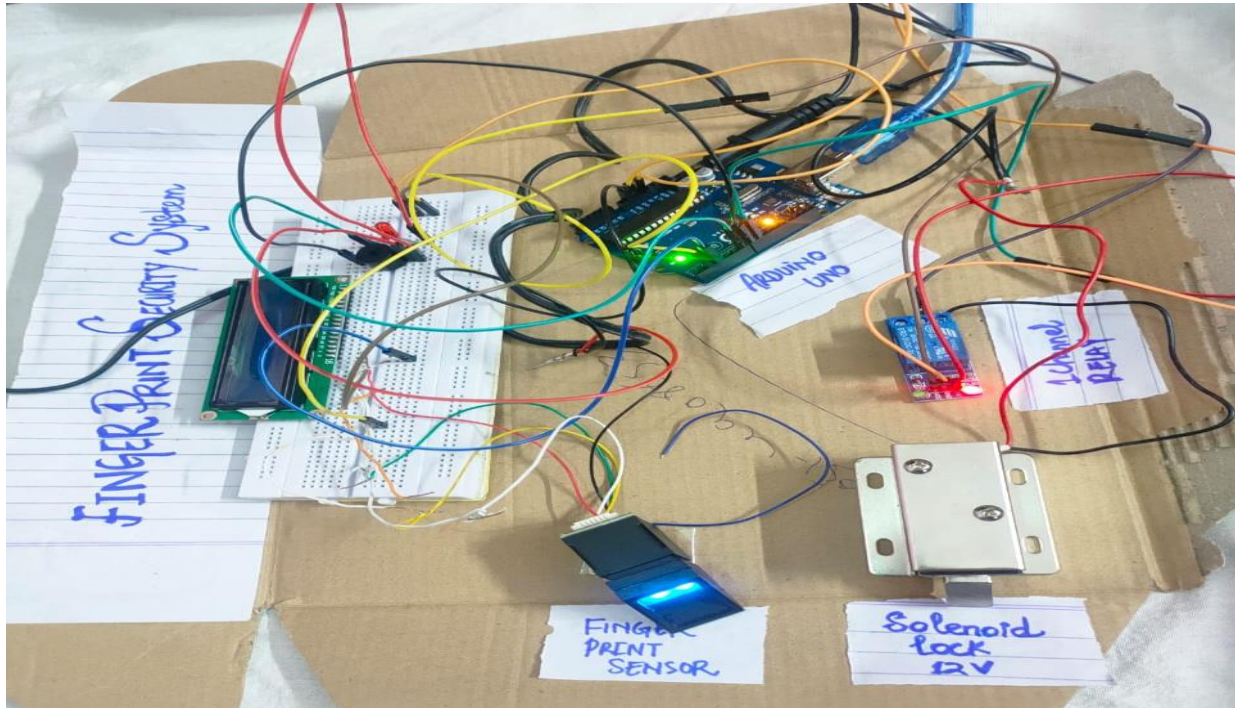
- **HARDWARE IMPLEMENTATION**

ACCESS GRANTED



As here you can see that in this case access is granted and Arduino has send the signal to one channel relay module to switch from normally open mode to normally close mode and then finally 12V voltage supply will reached at our solenoid lock and our door will open.

ACCESS DENIED



In this case an invalid entry tried to access our secure home at output our Buzzer will beep and door lock will remain close

DELETE CODE: -

```
void loop()                                // run over and over again
{
    Serial.println("Please type in the ID # (from 1 to 127) you want to delete...");
    uint8_t id = readnumber();
    if (id == 0) { // ID #0 not allowed, try again!
        return;
    }

    Serial.print("Deleting ID #");
    Serial.println(id);

    deleteFingerprint(id);
}

uint8_t deleteFingerprint(uint8_t id) {
    uint8_t p = -1;
```

```

p = finger.deleteModel(id);

if (p == FINGERPRINT_OK) {
    Serial.println("Deleted!");
} else if (p == FINGERPRINT_PACKETRECEIVEERR) {
    Serial.println("Communication error");
} else if (p == FINGERPRINT_BADLOCATION) {
    Serial.println("Could not delete in that location");
} else if (p == FINGERPRINT_FLASHERR) {
    Serial.println("Error writing to flash");
} else {
    Serial.print("Unknown error: 0x"); Serial.println(p, HEX);
}

return p;
}

```

Message (Enter to send message to 'Arduino Uno' on 'COM5')

```

Deleting ID #1
Deleted!
Please type in the ID # (from 1 to 127) you want to delete...
Deleting ID #26
Deleted!
Please type in the ID # (from 1 to 127) you want to delete...

```

We can delete any entry that we have stored for example we have stored a fingerprint with id #26 now we can delete this id through this delete code.

CONCLUSION

Fingerprint door locks are great investment for home or business. It provides great security by providing restrictions to unwanted access. This device increases level of security by adding unique biological features of authorized person. For anyone who wants more security to their homes, fingerprint based door lock system are best choice.